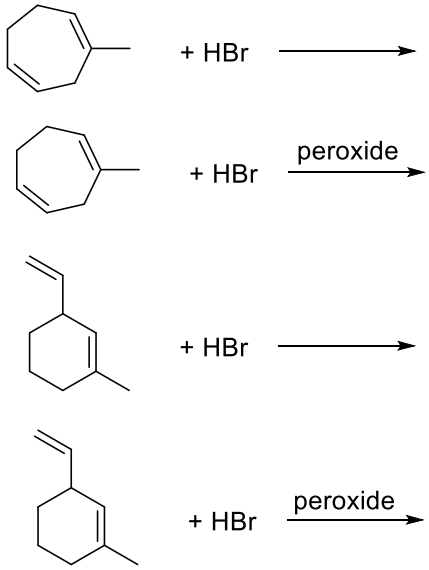
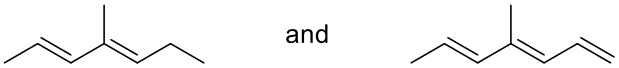
Organic Cumuluative Examination (60 points needed to pass)

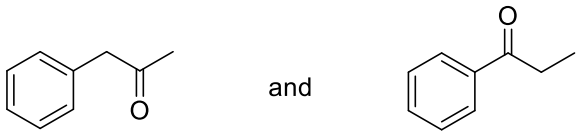
1) Predict the reaction product and complete the following reactions; one equivalent of each reagent is used (40 points)

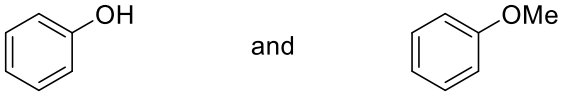
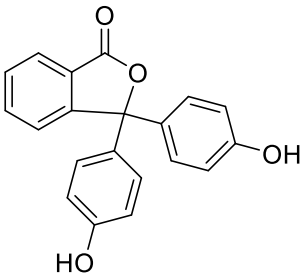


2) How could one use UV spectroscopy to distinguish between the compounds in each of the following pairs? (40 points)



CH2=CHCH=CHCH=CH2 and CH_2=CHCH=CHC(=O)CH_3





3) Phenolphthalein (structure below) is a compound that can change color change depending on the pH of the solution. Solutions of pH values smaller than 8.5 are colorless, and solutions with a pH larger than 8.5 are deep red-purple-pink. Explain the color change (10 points)